

Mérida Urban Intelligence

Integrating census, business, cartographic and public-safety data on one geography to measure, compare and explain the city, AGEB by AGEB.

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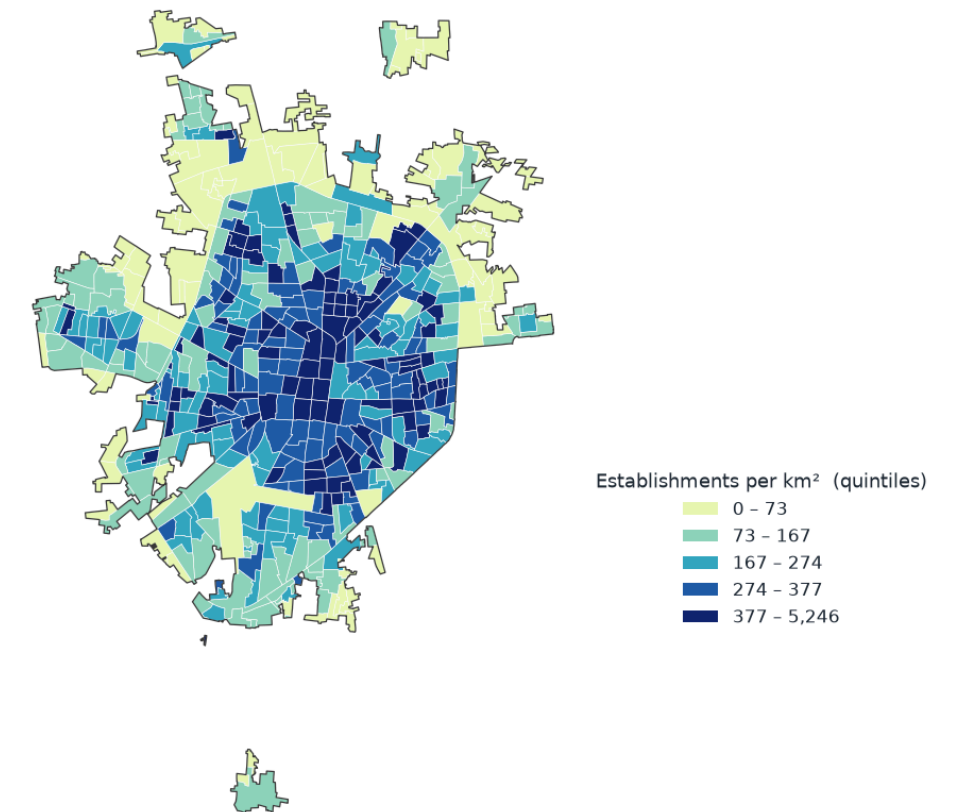
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Business density

Urban AGEBs of Mérida



Source: INEGI Census 2020, DENE and Marco Geoestadístico 2020 · computed in the warehouse

Four official sources, four incompatible geographies

Demographic

INEGI Census 2020
tables by block and AGEB

17,073 rows

Economic

INEGI DENU
points: one per establishment

**56,909
establishments**

Geographic

INEGI Marco
Geoestadístico 2020
polygons

526 urban AGEBs

Public safety

Georeferenced incidents
*points: not published for
Mérida*

pending

Goal: one warehouse in PostgreSQL/PostGIS that computes territorial KPIs, compares areas and tests whether demographic, economic and public-safety patterns are geographically associated.

Why the urban AGEB

Candidate	Units	Median population	PEA withheld	No business	
Block	17,595	52	4.8%	33%	too small
Urban AGEB	526	1,755	0.8%	3%	selected
Colonia	—	—	—	—	no census data
Locality	53	—	—	—	1 holds 483 AGEBs

526

urban AGEBs, keys match census 1:1

259.9 km²

study area

96.2%

of the municipal population covered

From latitude/longitude to AGEB

99.57%

of 56,909 establishments fall inside an AGEB polygon

99.82%

agree with the AGEB declared by INEGI — independent check

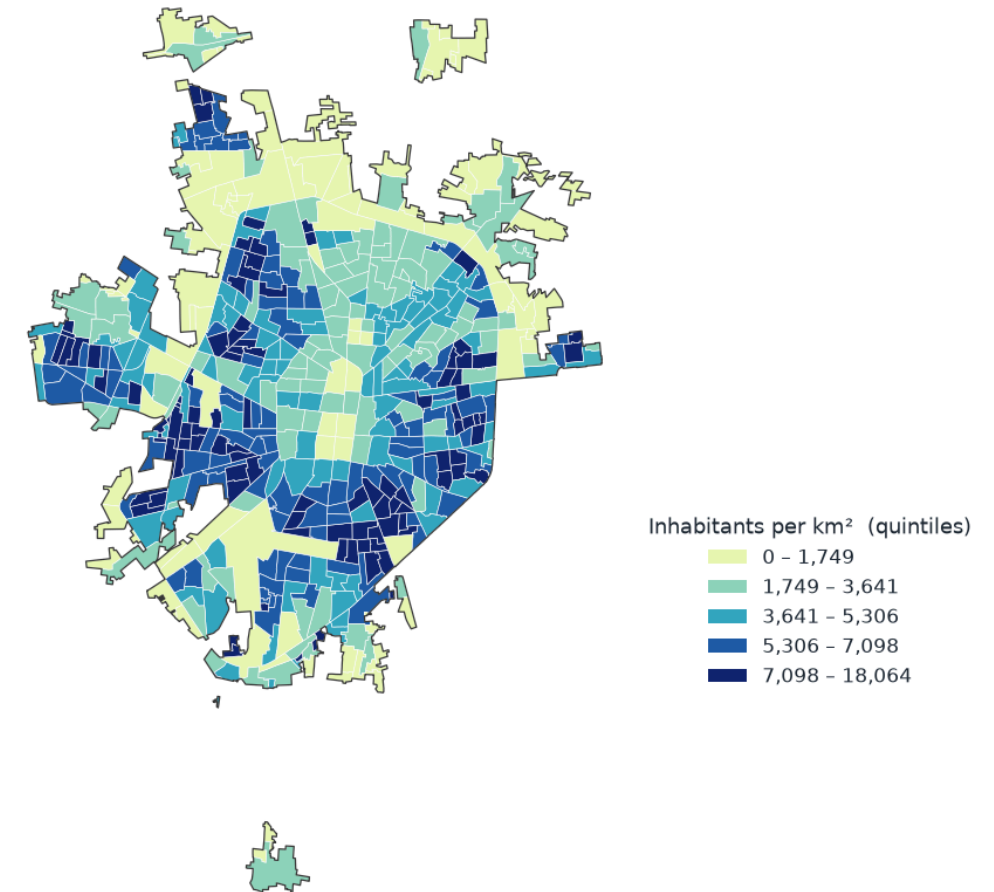
245

unmatched, 165 declared in rural areas; kept with a reason

- Points built in EPSG:4326 from WGS84 coordinates
- Rule: point within polygon; edge points to the lowest key
- Cartography has no EPSG code → parameters verified equal to **EPSG:6372**; areas measured there

Population density

Urban AGEBs of Mérida



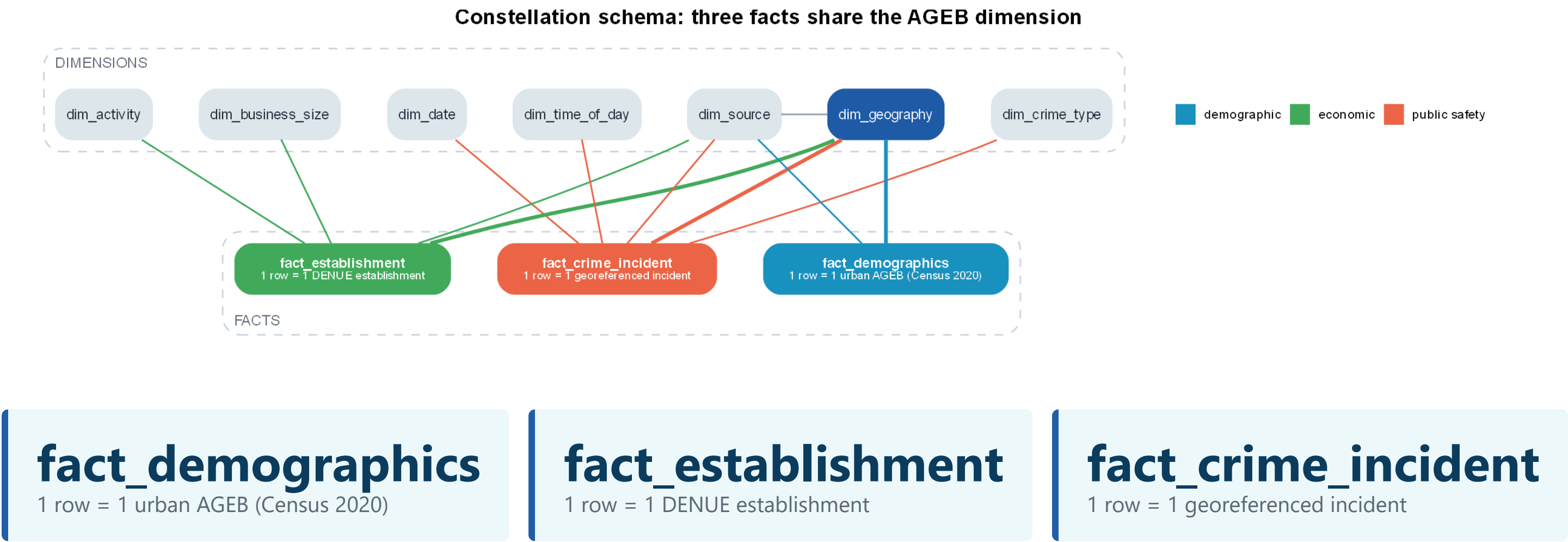
Source: INEGI Census 2020, DENUE and Marco Geoestadístico 2020 · computed in the warehouse

Reproducible from the original sources



Grain change made explicit: the census keeps INEGI's own AGEb totals instead of summing blocks, because blocks withheld for confidentiality would make the sums fall short.

Constellation schema on PostgreSQL/PostGIS



The 14 KPIs are SQL views on the warehouse

957,399

total population

3,684

inhabitants per km²

63.2%

economically active rate

9.8%

aged 65 and over

56,664

establishments

218

establishments per km²

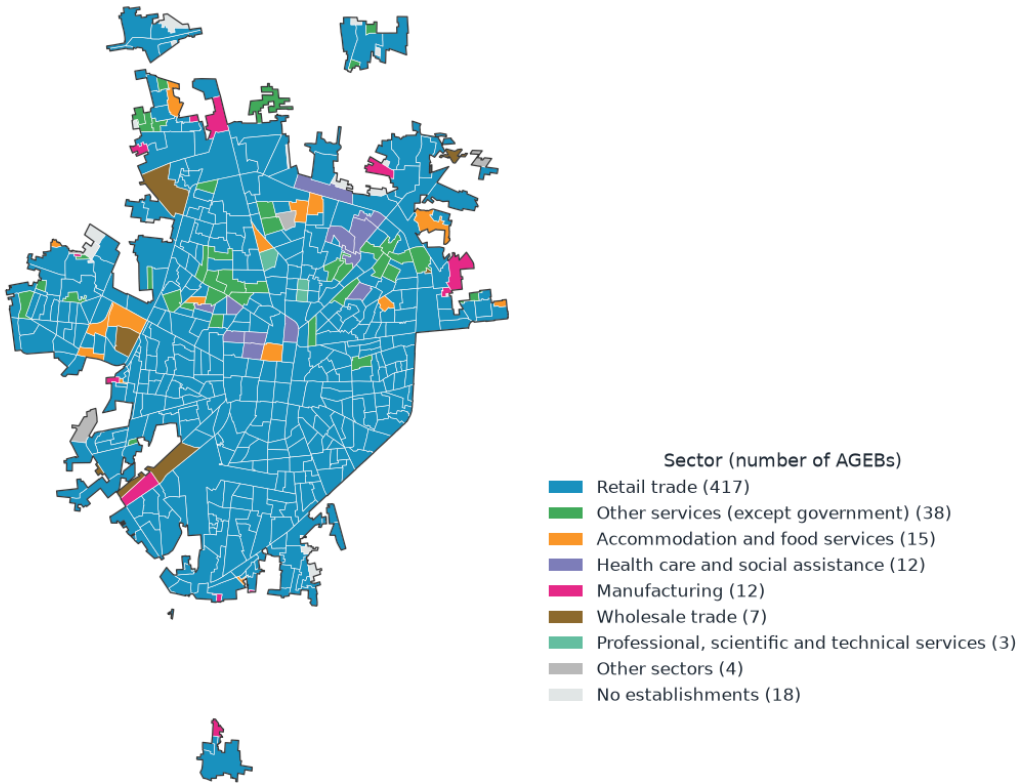
59.2

businesses per 1,000 residents

417

AGEBs dominated by retail

Dominant economic activity
Sector with the most establishments in each AGEB



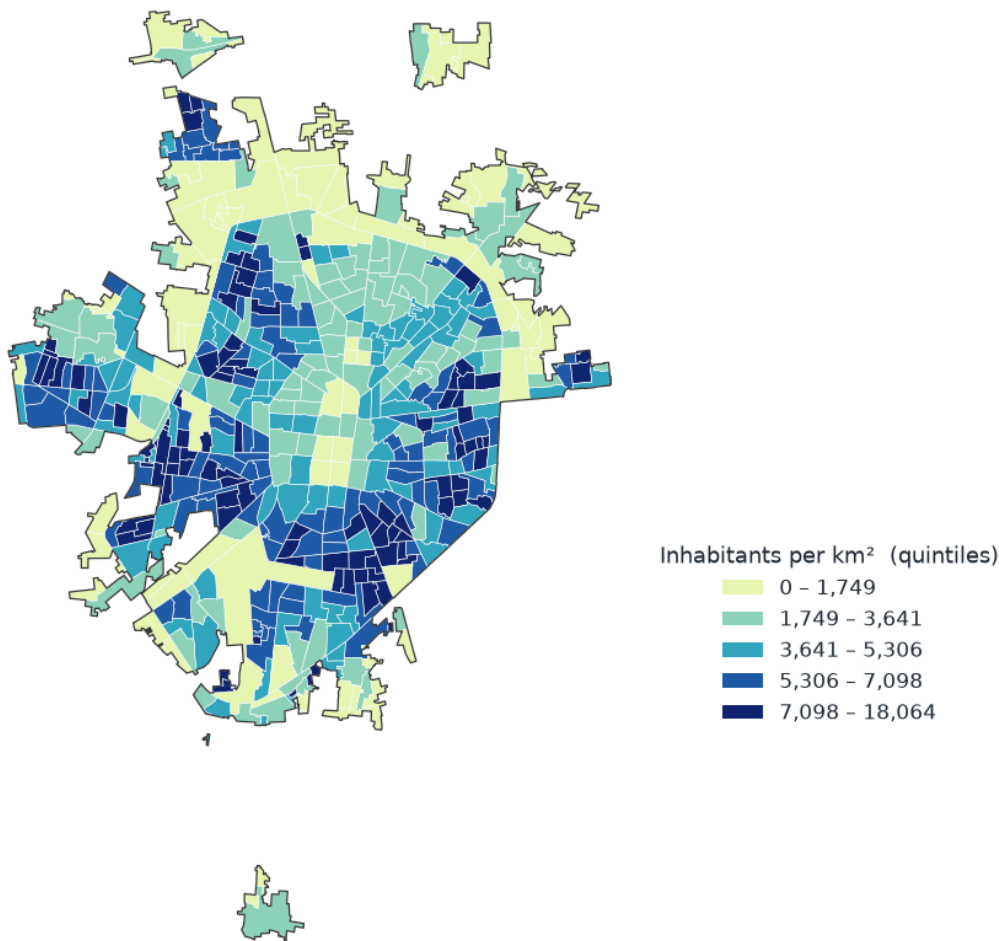
Source: INEGI Census 2020, DENUE and Marco Geoestadístico 2020 · computed in the warehouse

Rules: densities per km² · rates in % · NULL instead of zero when a denominator is missing · 32 AGEBs with fewer than 100 residents flagged for per-capita KPIs · crime KPIs empty until a georeferenced source exists.

Business concentrates in the centre; residents live around it

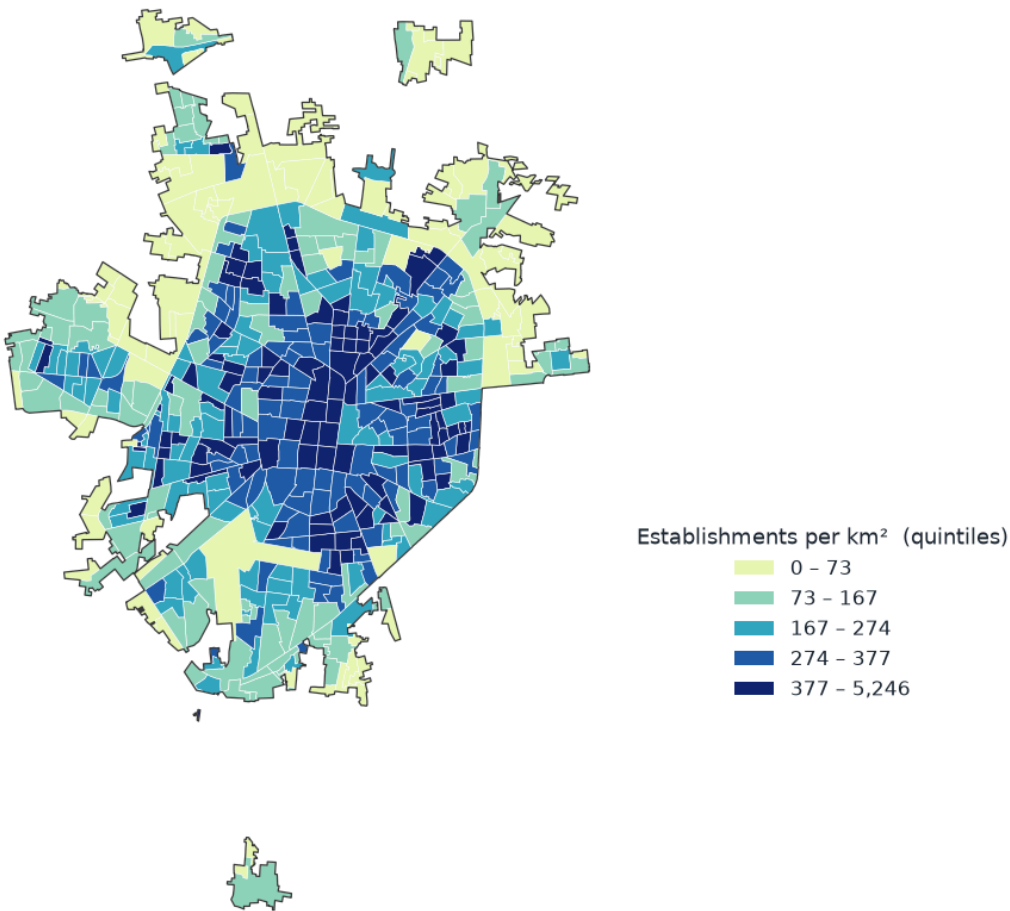
Population density

Urban AGEBs of Mérida



Business density

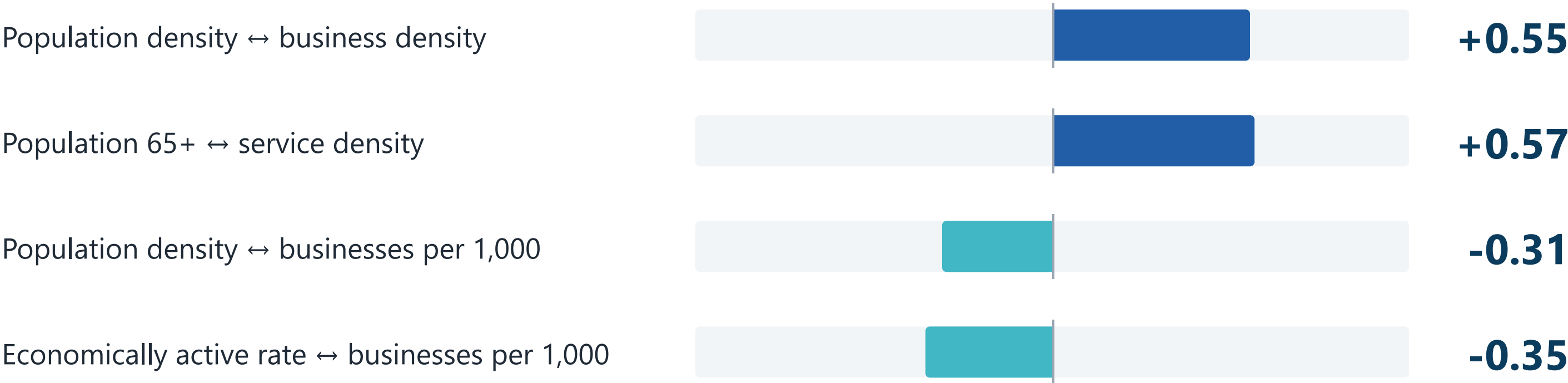
Urban AGEBs of Mérida



Source: INEGI Census 2020, DENU and Marco Geoestadístico 2020 · computed in the warehouse

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Denser areas have more businesses, but fewer per resident



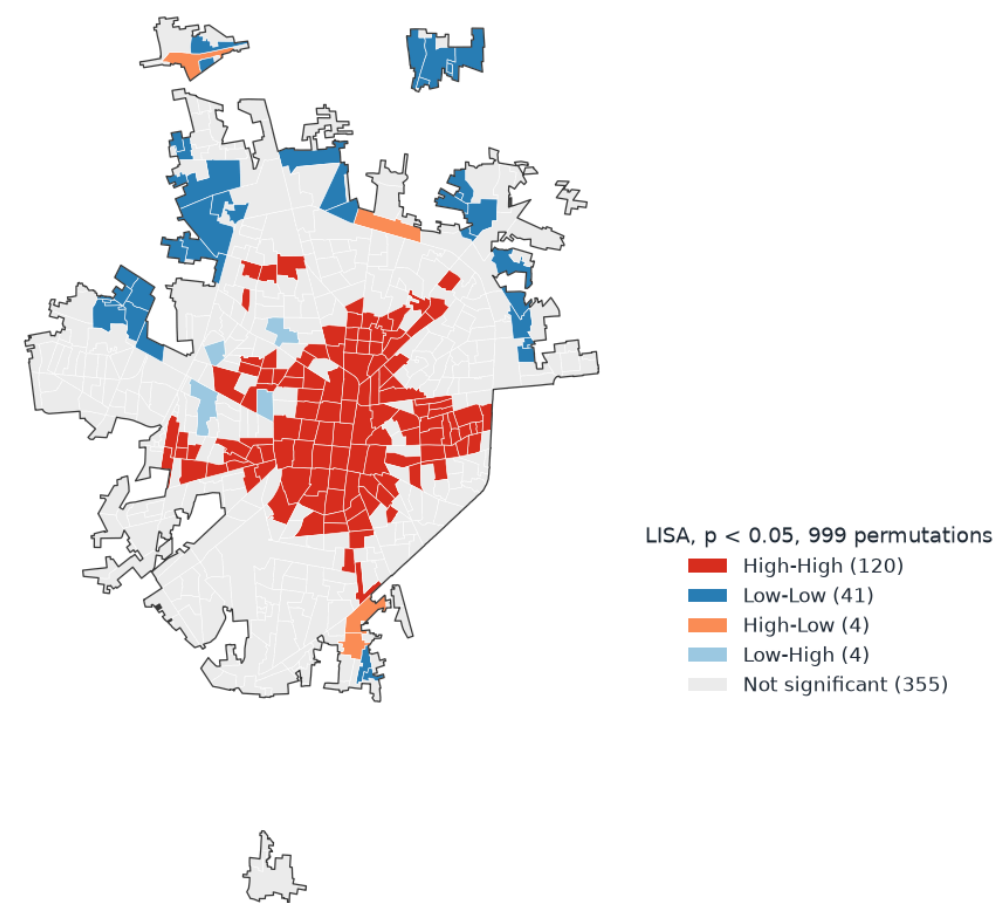
Spearman's ρ (skewed indicators), all $p < 0.001$, $n = 491\text{--}526$ AGEs; Pearson on log values agrees in sign. Newer dormitory neighbourhoods: many working-age residents, few local businesses.

Every indicator clusters in space

Indicator	Moran's I	p
Population density	0.37	0.001
Business density	0.51	0.001
Economically active rate	0.55	0.001
Businesses per 1,000	0.46	0.001

- Queen contiguity, row-standardised, log values
- 999 permutations; 2 detached AGEBs excluded
- LISA: **120** High-High AGEBs in the centre, **41** Low-Low on the northern edge

Local clusters of business density
Local Moran's I (LISA), Queen contiguity, log values



Source: INEGI Census 2020, DENUE and Marco Geoestadístico 2020 · computed in the warehouse

Business sits next to populated areas

0.32

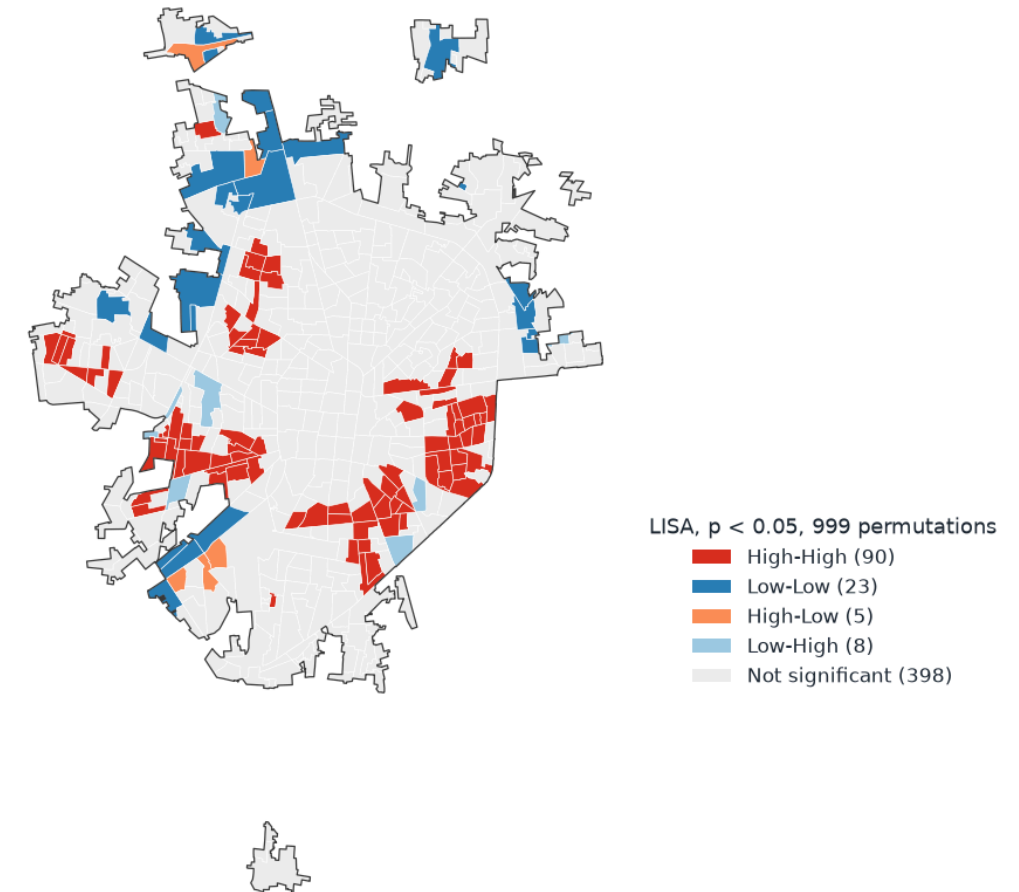
bivariate Moran's I, business density vs neighbouring population density ($p = 0.001$)

Public safety

No incident coordinates are published for Mérida (official data stop at municipal level). The crime path is built and tested; crime KPIs stay NULL — not zero — until a georeferenced source exists.

Business density vs neighbouring population density

Bivariate LISA: value of x in the AGEB against the mean of y in its neighbours



Source: INEGI Census 2020, DENUÉ and Marco Geoestadístico 2020 · computed in the warehouse

What the warehouse shows, and what it cannot

Findings

- Economic activity is concentrated in the historic centre and corridors; residents live in a ring around it.
- Per resident, the densest neighbourhoods have the fewest businesses: dormitory areas.
- All indicators cluster in space; local clusters identify where to focus planning.
- Older areas concentrate services.

Cautions

- Association is not causation; spatial dependence inflates ordinary p-values.
- Results describe AGEBs, not people (ecological fallacy, MAUP).
- Census 2020 vs current DENU; informal activity under-represented.
- Crime cannot be analysed inside the city without incident coordinates.